

EV Charging Station Network Management System

You are asked to develop a system to manage an electric vehicle charging station network, handle reservations and payments, and track charging sessions. The system serves EV drivers, station operators, and administrators.

Key Features

Station Catalog & Charger Types:

Maintains a database of charging stations including location details, charger types (AC/DC), power output (22kW, 50kW, 150kW), pricing per kWh, connector types (Type 2, CCS, CHAdeMO), and operating hours.

Interactive Map & Station Finder:

Integrates Google Maps API to display all stations on an interactive map. Users can view their current location, see nearby stations with real-time availability status (available/occupied/offline), and get directions to selected station.

Vehicle Registration:

Enables users to register their electric vehicles with details (brand, model, battery capacity, connector type, plate number). System validates vehicle-charger compatibility when making reservations.

Charging Reservation & Conflict Handling:

Enables users to select a station on the map and reserve a charging slot for a specific date and time. The system enforces rules (e.g., "Maximum 2 hours per session", "Cannot reserve more than 24 hours in advance") and prevents double-booking.

Online Payment & Wallet System:

Integrates a secure payment gateway for wallet top-ups. Charging fees are automatically deducted based on kWh consumed. Handles refunds for cancelled reservations and generates digital receipts.

Charging Session Tracking:

Records active charging sessions with real-time progress (current kWh, estimated time remaining, cost so far). Users can monitor their session status from the app.

Station Maintenance & Issue Reporting:

Users can report station issues (e.g., "Connector damaged", "Payment terminal broken"). Operators can mark chargers as out-of-service, automatically cancelling affected reservations and notifying users.

Favorites & Charging History:

Users can save favorite stations for quick access and view their complete charging history with session details, energy consumed, and costs.

Notifications:

System alerts for reservation confirmations, charging started/completed, low wallet balance, and

station availability updates.

Administrative Reporting:

Provides revenue reports, station utilization statistics, peak hour analysis, and user activity summaries.

Required Implementation

Students are required to implement the **vehicle registration module** and the **core reservation workflow**, which consists of selecting a station on the map, checking charger-vehicle compatibility, creating a reservation for a specific time slot, displaying the route to the selected charger using Google Maps integration, and processing the charging session to calculate the final cost.

Google Maps Integration

For API documentation and setup, visit: <https://mapsplatform.google.com/lp/maps-apis/>

Feature	Description
Location Display	Shows user's current location on the map
Station Markers	Displays all charging stations with markers
Color Codes	Green: Available, Yellow: Occupied, Red: Offline
Filtering	Filter by connector type, power output, price
Distance Calculation	Shows distance from user to each station
Navigation	Provides directions to selected station

User Flow Example

1. VEHICLE REGISTRATION

Brand:	Tesla	Battery:	75 kWh
Model:	Model 3	Connector:	CCS
Plate:	35 EV 2024		

2. STATION SEARCH ON MAP

User Location: Alsancak	
● Karsiyaka Hub (2.3 km)	
● Bornova Station (4.1 km)	
● Buca Point (6.8 km)	
	Available Occupied Offline

3. CREATE RESERVATION

Station:	Karsiyaka Hub	Date:	January 15, 2026
Charger:	DC 50kW #03	Time:	14:00 - 16:00
Compatibility:	CCS Supported ✓		

4. CHARGING SESSION

Start:	20% (15 kWh)	Unit Price:	4 TL/kWh
End:	80% (60 kWh)		
Consumed:	45 kWh	TOTAL:	180 TL